

IT 462:
Big Data Systems
Course Project
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Diabetes Health Indicators

Phase 2

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1. Introduction

The number of people surveyed in the "CDC Diabetes Health Indicators" dataset is 253,680. Every row in the data set represents a person who has answered the survey. In the data set, there are 22 variables. They represent the lifestyles, health, and diabetes of people. The data set has been created using the survey done in 2015. The name of the survey from which the data set has been created is "BRFSS 2015 survey." This data set has been used to find the variables associated with diabetes.

The objective of Phase 2 was to transform the raw dataset into a clean, consistent, and analysis-ready format suitable for scalable processing in Apache Spark and for subsequent RDD, SQL, and machine learning tasks. After preprocessing, the final dataset contains 229,500 records.

```
scala> import org.apache.spark.sql.functions._
      | import org.apache.spark.sql.expressions.Window
      | import org.apache.spark.sql.types._
      | val path = "/Users/dalalalyousef/Downloads/IT462_Project/data/diabetes_012_health_indicators_BRFSS2015.csv"
      |
      | val rawDF = spark.read
      |   .option("header", "true")
      |   .option("inferSchema", "true")
      |   .csv(path)
      |
      | println("Raw rows = " + rawDF.count())
Raw rows = 253680
import org.apache.spark.sql.functions._
import org.apache.spark.sql.expressions.Window
import org.apache.spark.sql.types._
val path: String = /Users/dalalalyousef/Downloads/IT462_Project/data/diabetes_012_health_indicators_BRFSS2015.csv
val rawDF: org.apache.spark.sql.DataFrame = [Diabetes_012: double, HighBP: double ... 20 more fields]
```

2. Data Cleaning

a. Missing Values

A full column-wise null value scan was conducted using Spark. A full column-wise null scan was conducted using Spark. No explicit null values were detected across any attributes. The dataset uses numeric survey coding (e.g., 0/1 values), meaning missing responses are not represented as SQL null values. Therefore, no imputation or removal was required for missing data.

Since no missing values were present, no imputation or row removal was required for this step.

```
scala> val nullCounts = rawDF.columns.map(c => rawDF.filter(col(c).isNull).count())
      | val nullSummary = rawDF.columns.zip(nullCounts)
      | nullSummary.foreach(println)
```

b. Duplicate Records

Duplicate rows were removed using Spark's `dropDuplicates()` operation.

- Rows before duplicate removal: 253,680
- Rows after duplicate removal: 229,781

A total of 23,899 duplicate records were identified and removed. Removing duplicates ensures that each observation represents a unique survey response and prevents bias in later analysis.

```
val dedupDF = rawDF.dropDuplicates()
println("After duplicate removal = " + dedupDF.count())
```

c. Range Validation and Outlier Handling

Logical range validation was applied to key numerical health indicators to ensure medical plausibility.

The following constraints were enforced:

- BMI: retained values between 10 and 80 [1]
- MentHlth: valid range 0–30 days
- PhysHlth: valid range 0–30 days

Records outside these ranges were considered invalid or extreme and were removed.

- Rows before validation: 229,781
- Rows after validation: 229,500
- Total rows removed due to invalid or extreme values: 281

This step improves dataset reliability by eliminating implausible survey responses that could distort statistical analysis or machine learning models.

```
val cleanedDF = dedupDF
  .filter(col("BMI") > 10 && col("BMI") < 80)
  .filter(col("MentHlth") >= 0 && col("MentHlth") <= 30)
  .filter(col("PhysHlth") >= 0 && col("PhysHlth") <= 30)
println("After range validation = " + cleanedDF.count())
```

3. Data Integration

Although the dataset consists of a single file, schema integration and standardization were performed to ensure consistency and readiness for analysis.

The following actions were applied:

- Renamed target variable Diabetes_012 to Diabetes
- Cast the Diabetes column to Integer type
- Verified and validated data types using Spark schema inspection

The final schema was consistent, standardized, and compatible with Spark SQL and ML pipelines.

```
scala> val integratedDF = cleanedDF
      |   .withColumnRenamed("Diabetes_012", "Diabetes")
      |   .withColumn("Diabetes", col("Diabetes").cast(IntegerType))
      |   integratedDF.printSchema()
```

4. Data Reduction

To reduce dimensionality and avoid redundancy, one attribute was removed:

- CholCheck was dropped from the dataset.

Justification:

CholCheck (whether cholesterol was checked in the last five years) represents screening behavior rather than an actual health condition. Since HighChol already captures cholesterol status, CholCheck was removed to reduce redundancy and keep the feature set focused on medical risk indicators.

- Number of attributes before reduction: 22
- Number of attributes after reduction: 21

This step reduces model complexity and potential multicollinearity.

```
scala> val reducedDF = integratedDF.drop("CholCheck")
      |   println("After feature reduction = " + reducedDF.columns.length)
```

5. Data Transformation

Feature transformation and engineering were applied to improve interpretability and analytical value.

a. BMI Category

BMI was converted from a continuous numerical variable into clinically meaningful categories:

- 0: Underweight (< 18.5)
- 1: Normal ($18.5\text{--}24.9$)
- 2: Overweight ($25\text{--}29.9$)
- 3: Obese (≥ 30)

This transformation enhances interpretability and aligns the dataset with medical classification standards.

b. UnhealthyLifestyleScore

A composite risk score was engineered to summarize unhealthy behavioral indicators:

UnhealthyLifestyleScore =

$$\text{Smoker} + \text{HvyAlcoholConsump} + (1 - \text{PhysActivity})$$

This score ranges from 0 to 3 and represents cumulative unhealthy lifestyle factors. It provides a compact representation of behavioral risk and reduces reliance on multiple correlated binary variables.

```
scala> val transformedDF = reducedDF
  |   .withColumn("BMI_Category",
  |     when(col("BMI") < 18.5, 0)
  |     .when(col("BMI") < 25, 1)
  |     .when(col("BMI") < 30, 2)
  |     .otherwise(3)
  |   )
  |   .withColumn("UnhealthyLifestyleScore",
  |     col("Smoker") +
  |     col("HvyAlcoholConsump") +
  |     (lit(1) - col("PhysActivity"))
  |   )
  |   println("Final rows = " + transformedDF.count())
Final rows = 229500
val transformedDF: org.apache.spark.sql.DataFrame = [Diabetes: int, HighBP: double ... 21 more fields]
```

6. References

- [1] Centers for Disease Control and Prevention, "About Adult BMI," 2023. [Online]. Available: <https://www.cdc.gov/bmi/adult-calculator/index.html>. [Accessed: Feb. 27, 2026].